Task 2: Feature Engineering, Model Optimization & Performance Comparison

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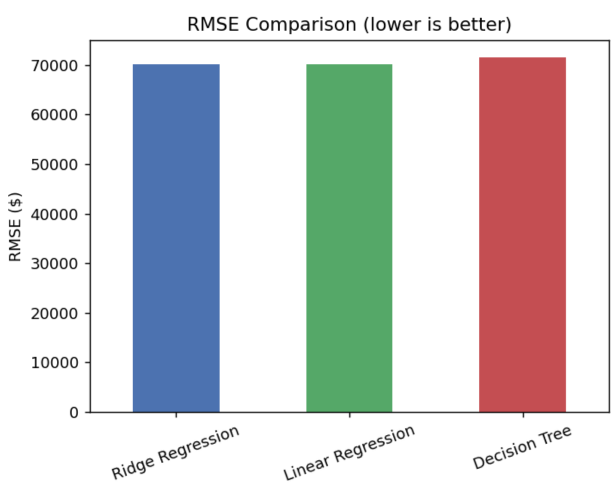
1. Methodology

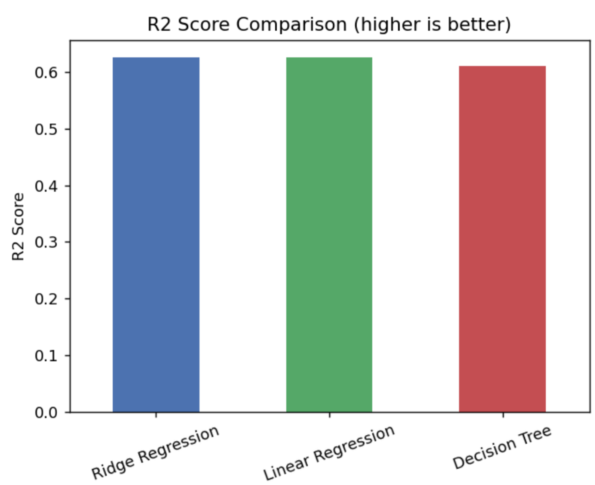
The California Housing dataset (20,640 rows) was used to build an enhanced House Price Prediction system. Missing values in total_bedrooms were imputed with the column median, and the categorical ocean_proximity feature was one-hot encoded. All features were then standardized using StandardScaler (mean 0, standard deviation 1) so that no single feature dominates model training purely due to its numeric scale — this is especially important for the regularized Ridge model. Data was split into 80% training and 20% testing sets (random_state=42). Three regression models were trained and evaluated on the same test set: Linear Regression (baseline), Ridge Regression (alpha=1.0, L2 regularization), and a Decision Tree Regressor (max_depth=5, to limit overfitting).

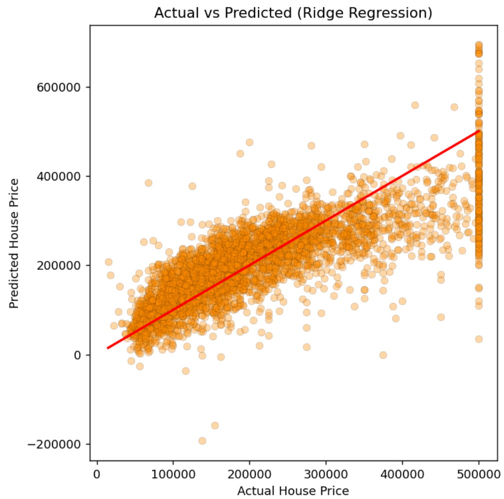
2. Model Performance Comparison

Model	MAE ($)	RMSE ($)	R^2 Score
Ridge Regression	50,668	70,057	0.6255
Linear Regression	50,671	70,061	0.6254
Decision Tree (depth=5)	50,341	71,511	0.6098

(Ridge Regression row highlighted — best performer on RMSE and R^2.)







Actual vs Predicted values for the best model (Ridge Regression). Points closer to the red line indicate more accurate predictions.

3. Results & Conclusion

Ridge Regression was selected as the best-performing model, achieving the lowest RMSE ($70,057) and highest R^2 (0.6255), marginally ahead of plain Linear Regression. This shows that L2 regularization gave a small stability improvement here without any loss in accuracy, which is the expected effect on a dataset that is not heavily overfit to begin with. The depth-limited Decision Tree had a competitive MAE but a noticeably higher RMSE, meaning it produces more large errors; a shallow tree cannot fully capture the underlying non-linear relationships in the data, while a deeper tree risks overfitting. Overall, $R^2 \approx 0.62$ across all models means about 62% of the variance in house prices is explained by the available features — a solid baseline that further feature engineering or ensemble methods (Random Forest, Gradient Boosting) could improve.

4. Deliverables

AI_ML_Task2_Model_Comparison.ipynb (full code, feature scaling, model training, comparison table, and plots), this PDF report, and the saved best model (best_model.joblib + scaler.joblib).